

Grade 8
Unit 8
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

Solve each equation, showing all your work.

1. $-2(n - 3) = -2n + 6$ $-2n + 6 = -2n + 6$ $-6 \quad -6$ $-\frac{2n}{-2} = -\frac{2n}{-2}$ $n = n$ Infinitely Many Solutions	2. $5y + 8 - 2y = 4y - y + 5$ $3y + 8 = 3y + 5$ $-3y \quad -3y$ $8 \neq 5$ No Solution
3. $4 - 3(j - 2) = 3(-j - 1)$ $4 - 3j + 6 = -3j - 3$ $+3j \quad +3j$ $4 + 6 = -3$ $10 \neq 3$ No Solution	4. $3.5(2m + 8) = 9 + 3m - 1$ $7m + 28 = 3m + 8$ $-3m \quad -3m$ $4m + 28 = 8$ $-28 \quad -28$ $4m = -20$ $m = -5$
5. $(\frac{6h+1}{4} = 2h - 3) \times 4$ $6h + 1 = 8h - 12$ $+12 \quad +12$ $6h + 13 = 8h$ $-6h \quad -6h$ $\frac{13}{2} = \frac{2h}{2}$ $6\frac{1}{2} = h$	6. $\frac{2}{3}(12a + 9) = 2(4a + 3)$ $8a + 6 = 8a + 6$ $-6 \quad -6$ $\frac{8a}{8} = \frac{8a}{8}$ $a = a$ Infinitely Many Solutions

7. $(\frac{1}{2}w + 3 = \frac{3}{4}w - 4) \times 4$ $2w + 12 = 3w - 16$ $+16 \quad +16$ $2w + 28 = 3w$ $-2w \quad -2w$ $28 = w$	8. $0.6(3k + 1) - 0.1k = k + 1.8 + 0.7k$ $1.8k + 0.6 - 0.1k = 1.7k + 1.8$ $1.7k + 0.6 = 1.7k + 1.8$ $-1.7k \quad -1.7k$ $0.8 \neq 1.8$ No Solutions
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9. For each of the equations in problems 1-8, determine whether it has one solution, no solutions, or infinitely many solutions. Write each equation in the appropriate box.

One Solution	No Solutions	Infinitely Many Solutions
#4: $3.5(2m + 8) = 9 + 3m - 1$ #5: $\frac{6h+1}{4} = 2h - 3$ #7: $\frac{1}{2}w + 3 = \frac{3}{4}w - 4$	#2: $5y + 8 - 2y = 4y - y + 5$ #3: $4 - 3(j - 2) = 3(-j - 1)$ #8: $0.6(3k + 1) - 0.1k = k + 1.8 + 0.7k$	#1: $-2(n - 3) = -2n + 6$ #6: $\frac{2}{3}(12a + 9) = 2(4a + 3)$

10. Soledad solved the equation $3(2g - 15) = 5g + g - 45$ and claimed that there are no solutions to this equation. Her work is shown below. Explain whether you agree with Soledad.

$$3(2g - 15) = 5g + g - 45$$

$$6g - 15 = 5g + g - 45$$

$$6g - 15 = 6g - 45$$

$$-15 = -45$$

No Solution

Explanations may vary.
I do not agree with Soledad.
Soledad did not distribute the 3 properly. If she had, she would have gotten $6g - 45 = 5g + g - 45$.
The correct solution is there are an infinite number of solutions.